Image-Based Visual Servoing for High-Speed Multicopter Interception via Safe Reinforcement Learning

A Staged Curriculum from Kinematic Validation to Control-Barrier-Function Safety with In-Policy Projection

Project Master Technical Report

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Abstract

We present an end-to-end study of training a reinforcement-learning policy for image-based visual servoing (IBVS) interception of a maneuvering aerial target, progressing through a nine-step curriculum from a kinematic pipeline-validation stage to a full six-degree-of-freedom (6-DOF) multicopter with realistic perception, environmental wind, and control-barrier-function (CBF) safety enforcement. The contribution is twofold. First, a *reproducible staged methodology*: each stage isolates a single new source of difficulty (action semantics, observation realism, sensor noise/delay, attitude dynamics, safety, disturbance), warm-started from the previous stage, with deterministic 200-episode evaluation at a fixed seed. Second, a *rigorous safe-RL ablation*: we compare an external CBF-QP safety filter against an in-policy differentiable CBF projection ("HardNet"), and through a controlled multi-seed study (variance baseline → entropy/curriculum interventions → auxiliary feasibility loss → λ ablation) we show that the in-policy projection, once its gradients are properly conditioned, attains the best worst-case robustness (36.7% success under ∼92% detection dropout and 1.5 m/s wind) while preserving nominal performance (∼91%) and hard attitude-safety guarantees ($\leq$ 0.02% limit-exceedance across thousands of episodes). Every numeric result is taken from committed evaluation logs.

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1 Introduction and Problem Formulation

1.1 Motivation

High-speed interception of an agile aerial target using only on-board vision is a canonical safety-critical control problem: the interceptor must keep the target inside its camera field of view (FOV) while closing distance rapidly, all under partial observability, sensor noise, and actuator limits. We adopt the image-based visual-servoing (IBVS) formulation of Yang et al. (arXiv:2404.08296), in which feedback is defined directly on the image plane, and we learn the guidance policy with Proximal Policy Optimization (PPO).

1.2 Coordinate Conventions and Notation

We use the North-East-Down (NED) earth-fixed coordinate system (EFCS), with z pointing down. The camera frame has x right, y down, z forward (optical axis). Key symbols:

- $\bar{\mathbf{p}} = (\bar{p}_x, \bar{p}_y)$ — normalized image coordinates of the target.
- $\mathbf{L}_s \in \mathbb{R}^{2\times 6}$ — IBVS interaction matrix (image Jacobian) relating camera spatial velocity to image-feature velocity.
- $\mathbf{u} = (a_x, a_y, a_z, \dot{\psi})$ — 4-D action: body-frame acceleration commands and yaw rate, normalized to $[-1, 1]$.
- $h_i(\mathbf{x})$ — CBF barrier functions (FOV-horizontal, FOV-vertical, pitch, roll).

1.3 The Staged Curriculum

Table 1 summarizes the curriculum. Each row introduces exactly one new difficulty, enabling clean attribution of performance changes. Figure 1 plots the headline success rate across stages.

Table 1: Staged curriculum. "Warm" = warm-started from the previous stage.

Stage	New difficulty	Train from	Headline success
1a	Pipeline validation (kinematic, full obs)	fresh	100%
1b	Acceleration control + action smoothing	warm	>90%
2a	Vision-only observation	fresh	∼70%
2b	Noise + delay + DKF observer	warm	62%
3a	First-order dynamics + attitude coupling	fresh	66% (mild)
3b	Full 6-DOF + SO(3) PD attitude control	warm	**96.5%** (mild)
4a	CBF safety (filter *and* in-policy)	fine-tune	89–92%
4b	Realistic perception + wind (DR)	warm	91% nominal

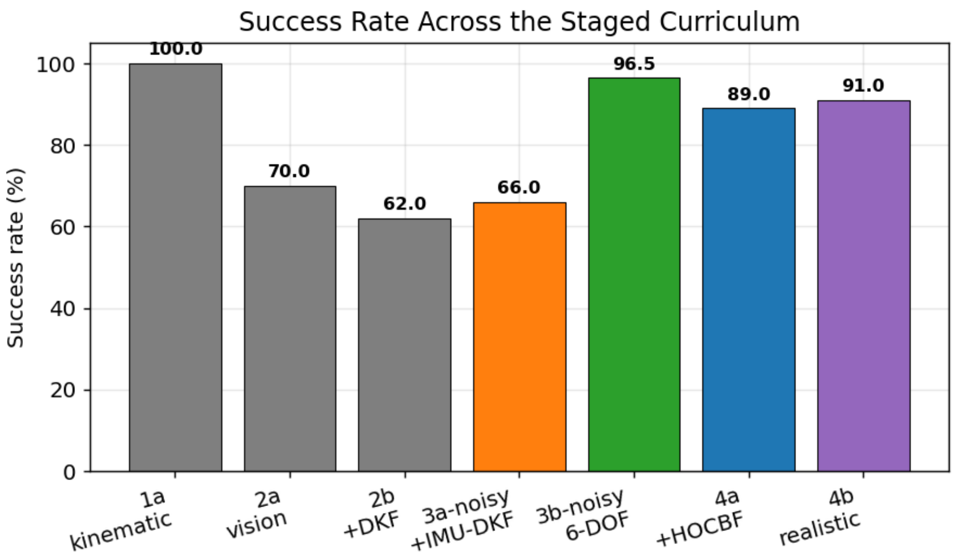


Figure 1: Success rate along the staged curriculum. Grey = early kinematic stages; orange = noisy kinematic; green = 6-DOF; blue = safety; purple = realistic perception. The 3a→3b jump is the dynamics-fidelity effect (§7).

1.4 Common Experimental Infrastructure

Algorithm: PPO (Stable-Baselines3), MLP policy. **Vectorization:** 16 parallel environments. **Rollout:** $n_{\text{steps}} = 2048$, batch 64, 10 epochs, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clip 0.2. **Timestep:** $dt = 20\,\text{ms}$ (50 Hz). **Evaluation:** deterministic policy, 200 episodes at fixed eval seed 1000 unless noted. **Late-stage:** linear learning-rate decay $3 \times 10^{-4} \to 0$ to mitigate the overtraining pattern observed throughout (best checkpoint typically precedes the final one).

2 Mathematical Formulations

This section gives the complete, implementation-faithful mathematical specification of every model, filter, controller, and safety operator used in the project. The stage chapters (§6–§12) reference these formulations and report experimental validation. All equations below correspond directly to the committed source (`models/`, `observers/`, `safety/`).

2.1 System State and the Pinhole Measurement Model

Let the earth-fixed (NED) position of interceptor and target be $\mathbf{p}, \mathbf{p}_t \in \mathbb{R}^3$ with z down. The relative position is $\mathbf{p}_r = \mathbf{p} - \mathbf{p}_t$ and the range $d = \|\mathbf{p}_r\|$. Let $R_b^e \in SO(3)$ be the body-to-earth rotation and R_c^b the fixed body-to-camera rotation (a $-\pi/2$ pitch placing the optical axis along body $+x$). A target point in the camera frame is

$$\mathbf{p}^c = R_c^b \, (R_b^e)^\top \, (\mathbf{p}_t - \mathbf{p}) = [x_c,\, y_c,\, z_c]^\top. \tag{1}$$

The normalized pinhole projection and field-of-view (FOV) test are

$$\bar{\mathbf{p}} = \left[\tfrac{x_c}{z_c},\, \tfrac{y_c}{z_c}\right]^\top, \qquad \text{in-FOV} \;\Leftrightarrow\; \left|\arctan \tfrac{x_c}{z_c}\right| \le \tfrac{\alpha_h}{2} \;\wedge\; \left|\arctan \tfrac{y_c}{z_c}\right| \le \tfrac{\alpha_v}{2}, \tag{2}$$

with horizontal/vertical FOV $\alpha_h = 70°$, $\alpha_v = 55°$. The FOV margin $m_{\text{FOV}} = \min(1 - \frac{|\arctan(x_c/z_c)|}{\alpha_h/2}, 1 - \frac{|\arctan(y_c/z_c)|}{\alpha_v/2}) \in [0, 1]$ (1 = centered, 0 = edge).

2.2 Interaction Matrix (Image Jacobian)

For a point feature at $\bar{\mathbf{p}} = (\bar{p}_x, \bar{p}_y)$ and depth z_c, the IBVS interaction matrix $\mathbf{L}_s \in \mathbb{R}^{2\times 6}$ maps the camera spatial velocity $[\mathbf{v}_c; \boldsymbol{\omega}_c]$ to image-feature velocity $\dot{\bar{\mathbf{p}}} = \mathbf{L}_s[\mathbf{v}_c; \boldsymbol{\omega}_c]$:

$$\mathbf{L}_s = \begin{bmatrix} -\frac{1}{z_c} & 0 & \frac{\bar{p}_x}{z_c} & \bar{p}_x\bar{p}_y & -(1+\bar{p}_x^2) & \bar{p}_y \\ 0 & -\frac{1}{z_c} & \frac{\bar{p}_y}{z_c} & 1+\bar{p}_y^2 & -\bar{p}_x\bar{p}_y & -\bar{p}_x \end{bmatrix}. \tag{3}$$

The rotational sub-block (columns 4–6) is depth-independent; this property underlies both the depth estimator (§2.5) and the IMU-aware DKF prediction (§2.6).

2.3 Target Dynamics

The target is a point mass integrated at dt with one of four maneuver modes, $\dot{\mathbf{p}}_t = \mathbf{v}_t,\ \dot{\mathbf{v}}_t = \mathbf{a}_t$:

- **Constant velocity:** $\mathbf{a}_t = \mathbf{0}$.
- **Sinusoidal evasion:** lateral acceleration $\mathbf{a}_t = A\,\omega^2 \sin(\omega t + \phi)\,\hat{\mathbf{e}}_\perp$ on an axis $\hat{\mathbf{e}}_\perp$ perpendicular to the initial velocity, amplitude $A = 3\,\mathrm{m}$, frequency $\omega = 1.5\,\mathrm{rad/s}$.
- **Circular:** orbit of radius $R = 15\,\mathrm{m}$ at rate $\omega = 0.5\,\mathrm{rad/s}$, $\mathbf{a}_t = -\omega^2(\mathbf{p}_t - \mathbf{c})$.
- **Random aggressive:** $\mathbf{a}_t$ resampled uniformly in $\|\mathbf{a}_t\| \le a_t^{\max}$ every 25 steps.

The target is observed only through its projection (2); no target state enters the policy observation directly (it appears only via $\bar{\mathbf{p}}$ and, during *training*, via the privileged range d in the reward).

2.4 Interceptor Dynamics — Two Models

(A) Kinematic first-order model (Stages 1–3a). The action $\mathbf{u} = (a_x, a_y, a_z, \dot{\psi})$ is a body-frame acceleration command plus yaw rate. Body velocity follows a first-order lag toward the commanded velocity $\mathbf{v}^\star = \mathrm{clip}(\mathbf{v}_b + \mathbf{a}\,dt, v_{\max})$:

$$\mathbf{v}_b \leftarrow \mathbf{v}_b + \alpha_v(\mathbf{v}^\star - \mathbf{v}_b), \qquad \alpha_v = \min(dt/\tau_v, 1),\ \tau_v = 0.15\,\mathrm{s}, \tag{4}$$

$$\theta_{\mathrm{des}} = -\arctan\tfrac{a_x}{g}, \quad \phi_{\mathrm{des}} = \arctan\tfrac{a_y}{g}, \qquad \theta \leftarrow \theta + \alpha_\theta(\theta_{\mathrm{des}} - \theta),\ \tau_\theta = 0.3\,\tau_v, \tag{5}$$

$$\dot{\psi} \leftarrow \dot{\psi} + \alpha_\psi(\dot{\psi}_{\mathrm{cmd}} - \dot{\psi}), \qquad \psi \leftarrow \psi + \dot{\psi}\,dt,\ \tau_\psi = 0.10\,\mathrm{s}, \tag{6}$$

with attitude limits $|\theta|, |\phi| \le 35^\circ$. The EFCS velocity is $\mathbf{v} = R_b^e\mathbf{v}_b$ and $\mathbf{p} \leftarrow \mathbf{p} + \mathbf{v}\,dt$. The defining feature is the acceleration–attitude coupling $\theta_{\mathrm{des}} = -\arctan(a_x/g)$: aggressive forward acceleration pitches the nose down, tilting the camera and threatening FOV loss.

(B) Full 6-DOF Newton–Euler model (Stages 3b–4b). The state is $(\mathbf{p}, \mathbf{v}, R_b^e, \boldsymbol{\omega}_b, f)$ with EFCS position /velocity, attitude, body angular rate, and scalar thrust. With mass m, gravity $\mathbf{g} = [0, 0, g]^\top$, and thrust acting along body $-z$:

$$\dot{\mathbf{p}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{g} - \tfrac{f}{m}R_b^e\mathbf{e}_3, \tag{7}$$

$$\dot{R}_b^e = R_b^e\,[\boldsymbol{\omega}_b]_\times, \qquad [\boldsymbol{\omega}_b]_\times = \mathrm{skew}(\boldsymbol{\omega}_b), \tag{8}$$

integrated by composing R_b^e with the incremental rotation $\exp([\boldsymbol{\omega}_b\,dt]_\times)$ (via a quaternion, avoiding Euler singularities). The motor and rate loops are first-order:

$$f \leftarrow f + \tfrac{dt}{\tau_m}(f_d - f), \qquad \boldsymbol{\omega}_b \leftarrow \boldsymbol{\omega}_b + \tfrac{dt}{\tau_r}(\boldsymbol{\omega}_d - \boldsymbol{\omega}_b), \quad \tau_m = \tau_r = 50\,\mathrm{ms}, \tag{9}$$

with $\boldsymbol{\omega}_b$ saturated to $\omega_{\max} = 4\,\mathrm{rad/s}$ and $f \in [0, f_{\max}]$, $f_{\max} = 3mg$ (thrust-to-weight 3). The desired thrust f_d and body rate $\boldsymbol{\omega}_d$ come from the SO(3) attitude controller (§3.1). Parameters: $m = 1\,\mathrm{kg}$, $J = \mathrm{diag}(0.01, 0.01, 0.02)\,\mathrm{kg\,m^2}$.

2.5 Depth Estimator (Inverse-Depth Kalman Filter)

Depth is estimated in inverse-depth coordinates $\rho = 1/z_c$, which linearizes the translational image-velocity relation. Subtracting the (depth-independent) rotational contribution from the measured image velocity gives the translational residual $\dot{\bar{\mathbf{p}}}_{\mathrm{tr}} = \dot{\bar{\mathbf{p}}} - \mathbf{L}_{\mathrm{rot}}\boldsymbol{\omega}_c$, and with $\mathbf{b} = \mathbf{B}(\bar{\mathbf{p}})\,\mathbf{v}^c_{\mathrm{rel}}$ (where $\mathbf{B} = [\,\mathbf{I}_2 \quad -\bar{\mathbf{p}}\,]$) the relation $\dot{\bar{\mathbf{p}}}_{\mathrm{tr}} = \rho\,\mathbf{b}$ is *linear* in ρ. A scalar Kalman filter then runs

$$\rho^- = \rho,\ \ P^- = P + Q; \quad K = \frac{P^-\mathbf{b}^\top}{\mathbf{b}P^-\mathbf{b}^\top + R},\ \ \rho \leftarrow \rho^- + K(\dot{\bar{\mathbf{p}}}_{\mathrm{tr}} - \rho^-\mathbf{b}),\ \ P \leftarrow (1 - K\mathbf{b})P^-, \tag{10}$$

with ρ clamped to $[\rho_{\min}, \rho_{\max}] = [0.005, 2]$ (i.e. $z_c \in [0.5, 200]$ m) and the update gated by an observability threshold $\|\mathbf{b}\|^2 > \epsilon$.

2.6 Disturbance Kalman Filter (Image-Plane State, Delay-Compensated)

The DKF estimates the current image-plane state $\mathbf{x} = [\bar{p}_x, \bar{p}_y, \dot{\bar{p}}_x, \dot{\bar{p}}_y]^\top$ from noisy, D-step-delayed measurements. Constant-velocity process and position-only measurement models:

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + \mathbf{w}_k, \quad F = \begin{bmatrix} \mathbf{I}_2 & dt\,\mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \qquad \mathbf{z}_k = H\mathbf{x}_{k-D} + \mathbf{n}_k, \quad H = [\mathbf{I}_2\ \ \mathbf{0}], \tag{11}$$

$\mathbf{w}_k \sim \mathcal{N}(0, Q)$, $\mathbf{n}_k \sim \mathcal{N}(0, R)$, $Q = \mathrm{diag}(\sigma_p^2, \sigma_p^2, \sigma_v^2, \sigma_v^2)$ with a deliberately large velocity process noise $\sigma_v = 0.5$ (disturbance-aware tracking of maneuvers without an explicit target model).

Prediction (standard): $\mathbf{x} \leftarrow F\mathbf{x}$, $P \leftarrow FPF^\top + Q$.

Prediction (IMU-aware upgrade). The camera ego-motion induces an image velocity $\dot{\bar{\mathbf{p}}}_{\mathrm{cam}} = \mathbf{L}_s[\mathbf{v}_c; \boldsymbol{\omega}_c]$ that changes *abruptly* during aggressive attitude changes — precisely when the constant-velocity assumption fails. We therefore feed forward only the *change* in the camera contribution, applying CV to the (slowly-varying) target residual:

$$\Delta_{\mathrm{cam}} = \dot{\bar{\mathbf{p}}}_{\mathrm{cam},k} - \dot{\bar{\mathbf{p}}}_{\mathrm{cam},k-1}, \quad \dot{\bar{\mathbf{p}}} \mathrel{+}= \Delta_{\mathrm{cam}}, \quad \bar{\mathbf{p}} \mathrel{+}= \tfrac{1}{2}dt\,\Delta_{\mathrm{cam}}. \tag{12}$$

Delay-compensated update. The measurement $\mathbf{z}_k$ pertains to time k–D. Using the buffered prediction $\mathbf{x}_{k-D}$ and covariance P_{k-D} for the innovation, the gain is applied to the *current* state (a small-D approximation):

$$\mathbf{y} = \mathbf{z}_k - H\mathbf{x}_{k-D},\ \ S = HP_{k-D}H^\top + R,\ \ K = P\,H^\top S^{-1},\ \ \mathbf{x} \leftarrow \mathbf{x} + K\mathbf{y}, \tag{13}$$

with the Joseph-form covariance update $P \leftarrow (\mathbf{I} - KH)P(\mathbf{I} - KH)^\top + KRK^\top$ for numerical stability. The 4-D DKF is the project's *perception ceiling*: it saturates between mild (D=1, σ=.015) and full (D=3, σ=.03) noise regardless of dynamics fidelity (§6,7).

3 Control and Safety Framework

3.1 SO(3) Geometric Attitude Controller

For the 6-DOF model, the policy's body-acceleration command $\mathbf{a}_b$ is realized by a geometric controller on $SO(3)$ (Lee et al. 2010). Lift the command to EFCS, $\mathbf{a}_d = R^e_b\mathbf{a}_b$, and form the required thrust force $\mathbf{f}_{\mathrm{req}} = m(\mathbf{a}_d - \mathbf{g})$. Its magnitude and direction set the desired thrust and tilt:

$$f_d = \mathrm{clip}(\|\mathbf{f}_{\mathrm{req}}\|, 0, f_{\max}), \qquad \hat{\mathbf{n}}_{f_d} = \mathbf{f}_{\mathrm{req}}/\|\mathbf{f}_{\mathrm{req}}\|. \tag{14}$$

The desired rotation $R_d = R_{\text{tilt}} R_b^e$ is the minimal tilt aligning the current thrust axis $-R_b^e \mathbf{e}_3$ with $\hat{\mathbf{n}}_{f_d}$ (Rodrigues' formula on the axis $\hat{\mathbf{n}}_{\text{cur}} \times \hat{\mathbf{n}}_{f_d}$). The attitude error and PD body-rate command are

$$\mathbf{e}_R = \tfrac{1}{2}\big(R_d^\top R_b^e - (R_b^e)^\top R_d\big)^\vee, \qquad \boldsymbol{\omega}_d = -k_{\text{att}}\,\mathbf{e}_R, \quad k_{\text{att}} = 6, \tag{15}$$

where $(\cdot)^\vee$ is the inverse hat (skew→vector) map. The yaw component is overridden by the policy's direct yaw-rate command, $\omega_{d,z} = \dot{\psi}_{\text{cmd}}$, and $\boldsymbol{\omega}_d$ is saturated to $\omega_{\max}$. The pair $(f_d, \boldsymbol{\omega}_d)$ then drives the first-order motor and rate loops of §2.4. This geometric controller produces markedly smoother attitude responses than the kinematic $\arctan(a/g)$ coupling, which is the mechanism behind the large 3a→3b gain.

3.2 Control Barrier Functions and the Safe Polytope

We enforce four safety constraints as zeroing CBFs $h_i(\mathbf{x}) \geq 0$ with the smooth squared form (nonzero gradient at the boundary, the regularity condition for forward invariance):

$$h_{\text{hfov}} = \tan^2\tfrac{\alpha_h}{2} - \bar{p}_x^2, \qquad h_{\text{vfov}} = \tan^2\tfrac{\alpha_v}{2} - \bar{p}_y^2, \tag{16}$$

$$h_{\text{pitch}} = \theta_{\max,\text{eff}}^2 - \theta^2, \qquad h_{\text{roll}} = \phi_{\max,\text{eff}}^2 - \phi^2, \tag{17}$$

where $\theta_{\max,\text{eff}} = \theta_{\max} - \delta_s$ uses an inner safety margin δ_s (default 0.10 rad) absorbing the proxy–truth mismatch below. The safe set is the intersection $\mathcal{C}(\mathbf{x}) = \{\, \mathbf{u} : A(\mathbf{x})\mathbf{u} \geq \mathbf{b}(\mathbf{x}),\ \mathbf{u} \in [-1,1]^4 \,\}$, a polytope. The continuous-time CBF condition $L_f h_i + L_g h_i\,\mathbf{u} + \alpha_i h_i \geq 0$ rearranges to the affine row $A_i \mathbf{u} \geq b_i$ with $A_i = L_g h_i$ and $b_i = -L_f h_i - \alpha_i h_i$.

Relative degree and the proxy linearization. In the true 6-DOF system the action (acceleration / yaw-rate) reaches the FOV and attitude outputs only through the attitude-tracking lag and Newton integration — a relative degree ≥ 2, for which $L_g h_i \equiv 0$ and the plain CBF-QP is ill-posed. We adopt a *control-affine proxy*: pitch and roll track $\theta_{\text{des}} = -a_x/g$, $\phi_{\text{des}} = a_y/g$ through a single τ_r lag, rendering all four constraints relative-degree-1 with closed-form Lie derivatives. For the attitude barriers, $\dot{\theta} = \frac{1}{\tau_r}(-a_x/g - \theta)$ gives

$$L_f h_{\text{pitch}} = \tfrac{2\theta^2}{\tau_r}, \qquad L_g h_{\text{pitch}} = \Big[\tfrac{2\theta}{g\tau_r}, 0, 0, 0\Big], \tag{18}$$

and symmetrically for roll (a_y, with sign flip). For the FOV barriers the $\mathbf{u}$-dependence flows through $\boldsymbol{\omega}_c = R_c^b \boldsymbol{\omega}_b$ with $\boldsymbol{\omega}_b$ a function of $(a_x, a_y, \dot{\psi})$; writing the per-step image velocity $\dot{\bar{\mathbf{p}}} = \mathbf{L}_s[\mathbf{v}_{\text{rel}}^c; \boldsymbol{\omega}_c]$ and differentiating $h_{\text{hfov}} = \tan^2(\cdot) - \bar{p}_x^2$,

$$L_f h_{\text{hfov}} = -2\bar{p}_x\,\dot{\bar{p}}_{x,\text{drift}}, \qquad L_g h_{\text{hfov}} = -2\bar{p}_x\,\big[\mathbf{L}_{\text{rot}} R_c^b M\big]_{0,:}, \tag{19}$$

where M is the constant Jacobian $\partial\boldsymbol{\omega}_b/\partial\mathbf{u}$ of the proxy. FOV rows are deactivated (A_i=0) when the target is out of frame. Because the proxy is first-order while the truth is second-order, the proxy responds *faster* than reality; the inner margin δ_s and a large class-$\mathcal{K}$ rate α absorb the mismatch, and the empirical attitude-limit exceedance stays $\leq 0.02\%$ of steps across thousands of episodes.

The CBF-QP (external filter, Stage 4a.3). Given the RL action $\mathbf{u}_{\text{RL}}$, the minimally-invasive safe action solves

$$\mathbf{u}_{\text{safe}} = \arg\min_{\mathbf{u}} \tfrac{1}{2}\|\mathbf{u} - \mathbf{u}_{\text{RL}}\|^2 \ \text{ s.t. }\ A\mathbf{u} \geq \mathbf{b},\ \mathbf{u} \in [-1,1]^4, \tag{20}$$

a 4-variable, ≤ 12-constraint QP solved with `quadprog` in ~ 0.15 ms. With a bisection fallback if (20) is infeasible (deep corners of the action box).

3.3 HardNet: Differentiable In-Policy Projection

HardNet replaces the external filter by embedding the projection onto $\mathcal{C}(\mathbf{x})$ *inside* the policy, so gradients propagate through it during PPO. The network emits $\mathbf{u}_{\text{raw}}$ and the policy mean is the Euclidean projection

$$\mathbf{u}_{\text{safe}} = \Pi_{\mathcal{C}(\mathbf{x})}(\mathbf{u}_{\text{raw}}) = \arg\min_{\mathbf{u}\in\mathcal{C}(\mathbf{x})} \tfrac{1}{2}\|\mathbf{u}-\mathbf{u}_{\text{raw}}\|^2. \tag{21}$$

Since $\mathcal{C}$ is an intersection of half-spaces and the box, we compute Π by **Dykstra's alternating projection**, which (unlike plain POCS) converges to the true Euclidean projection. With correction terms $\mathbf{p}^{(i)}, \mathbf{q}$ initialized to zero, each sweep projects onto every active half-space $\{\mathbf{u} : A_i\mathbf{u} \geq b_i\}$,

$$\mathbf{y} = \mathbf{u} + \mathbf{p}^{(i)}, \quad \mathbf{u}' = \mathbf{y} + \frac{\max(0,\, b_i - A_i\mathbf{y})}{\|A_i\|^2} A_i^\top, \quad \mathbf{p}^{(i)} \leftarrow \mathbf{y} - \mathbf{u}', \quad \mathbf{u} \leftarrow \mathbf{u}', \tag{22}$$

followed by a box projection $\mathbf{u} \leftarrow \text{clip}(\mathbf{u}+\mathbf{q}, -1, 1)$, $\mathbf{q} \leftarrow (\mathbf{u}+\mathbf{q}) - \mathbf{u}$, for $K{=}20$ sweeps. Each elementary projection is piecewise-linear and differentiable, so $\partial\mathbf{u}_{\text{safe}}/\partial\mathbf{u}_{\text{raw}}$ is the product of the active-face projectors: it equals $\mathbf{I}$ on the interior (gradient passes cleanly) and drops rank along active constraint normals (the gradient pathology motivating Intervention D, §9). The per-step coefficients $(A, \mathbf{b})$ are appended to the observation (16→36-D) by a context wrapper; a slicing feature-extractor keeps the network conditioned on only the 16-D base observation so weights remain interchangeable with the unconstrained MLP policy (enabling warm-start). Crucially, the executed and reward-relevant action is always $\mathbf{u}_{\text{safe}}$, so the projection can neither produce unsafe behavior nor (by itself) degrade task performance — it reshapes only the internal pre-image.

Intervention D — auxiliary feasibility loss. To keep the projection Jacobian full-rank (healthy gradients) the PPO objective is augmented with

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda\,\mathbb{E}\big[\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2\big], \tag{23}$$

which pulls $\mathbf{u}_{\text{raw}}$ toward feasibility so the projection approaches the identity. The weight λ trades a "lazy policy" (over-conservative at large λ) against gradient health; the ablation (§9) locates the Pareto frontier.

3.4 Reward Function (Canonical Form)

All stages share the reward template below; stage-specific weights and gating are noted in each chapter. With $\bar{\mathbf{p}}$ the image error, d the *privileged* range (training-only "teacher" signal, never in the observation), and $\mathbf{a}$ the normalized action:

$$\begin{aligned} r = {} & \underbrace{w_{\text{img}}\, e^{-k_1\|\bar{\mathbf{p}}\|^2} \max\!\big(0, 1-\tfrac{d}{d_{\text{cut}}}\big)\mathbb{1}_{\text{FOV}}}_{\text{distance-gated centering}} + \underbrace{w_{\text{app}}(d_{k-1}-d_k)}_{\text{approach}} + \underbrace{w_{\text{dist}}\tfrac{d}{d_{\text{norm}}}}_{\text{anti-loiter}} \\ & + \underbrace{w_{\text{brake}}\|\mathbf{v}_b\|\tfrac{(d_b-d)^+}{d_b}}_{\text{terminal braking}} + \underbrace{w_{\text{fov}}\mathbb{1}_{\neg\text{FOV}}}_{\text{FOV loss}} + \underbrace{w_{\text{bnd}}(1-m_{\text{FOV}})^2\mathbb{1}_{\text{FOV}}}_{\text{boundary}} \\ & + w_{\text{eff}}\|\mathbf{a}\|^2 + w_{\text{jerk}}\|\mathbf{a}-\mathbf{a}_{k-1}\|^2 + w_{\text{att}}\big(\tilde{\theta}^2+\tilde{\phi}^2\big) + w_{\text{time}} + w_{\text{int}}\mathbb{1}_{\text{success}}, \end{aligned} \tag{24}$$

with $\tilde{\theta} = \theta/\theta_{\max}$, $\tilde{\phi} = \phi/\phi_{\max}$, success when $d < d_{\text{succ}} = 2\,\text{m}$, and FOV-loss termination after 15 consecutive out-of-frame steps. The canonical (Stage 3+) weights are $w_{\text{img}}{=}0.4$, $w_{\text{app}}{=}2.0$, $w_{\text{dist}}{=}-0.15$, $w_{\text{brake}}{=}-0.05$ ($d_b{=}5\,\text{m}$), $w_{\text{fov}}{=}-3.0$, $w_{\text{bnd}}{=}-0.5$, $w_{\text{eff}}{=}-0.05$, $w_{\text{jerk}}{=}-0.05$, $w_{\text{att}}{=}-0.1$, $w_{\text{time}}{=}-0.05$, $w_{\text{int}}{=}200$, $k_1{=}5$, $d_{\text{cut}}{=}25\,\text{m}$, $d_{\text{norm}}{=}50\,\text{m}$. The *distance-gating* of the centering term ($\max(0, 1-d/d_{\text{cut}})$) and the anti-loiter penalty are the decisive fixes for the Stage 3a reward-hacking failure (§6).

3.5 Causal Chain (Per-Step Closed Loop)

Each 20 ms control step executes the following chain, which the stage chapters specialize:

1. **Sense:** camera projects the target (2); (Stage 2b+) measurement is noised and delayed by D steps; the intermittent detector (Stage 4b) may drop it.
2. **Estimate:** the DKF (2.6) fuses the delayed measurement with IMU-aware prediction to estimate the current image state; the inverse-depth filter estimates z_c.
3. **Observe:** the 16-D observation (image error/velocity, in-FOV flag, depth estimate, body velocity, Euler angles, previous action) is assembled; (HardNet) the CBF coefficients $(A, \mathbf{b})$ are appended ($\rightarrow$36-D).
4. **Decide:** PPO policy maps observation to $\mathbf{u}_{\text{raw}}$.
5. **Project (safety):** external CBF-QP (20) or in-policy Dykstra projection §3.3 yields $\mathbf{u}_{\text{safe}} \in \mathcal{C}$.
6. **Actuate:** (kinematic) first-order velocity/attitude lags, or (6-DOF) SO(3) controller (15) $\rightarrow$ motor/rate loops $\rightarrow$ Newton–Euler integration (7); (Stage 4b) wind drag $\mathbf{a}_{\text{drag}} = -k(\mathbf{v} - \mathbf{v}_{\text{wind}})$ is added.
7. **Reward & learn:** reward (24) is computed on the *executed* $\mathbf{u}_{\text{safe}}$; PPO updates the policy (and, under Intervention D, the feasibility loss conditions the projection gradients).

4 Stage 1a–1b: Pipeline Validation and Action Semantics

4.1 Theoretical & Algorithmic Foundations

Stage 1a validates the full Gymnasium+PPO pipeline with a kinematic interceptor (instant velocity commands) and a privileged 18-D observation that includes ground-truth relative distance, line-of-sight, and relative velocity. The reward combines an image-centering term $r_{\text{img}} = \exp(-k_1\|\bar{\mathbf{p}}\|^2)$, an approach term proportional to closure rate, and an interception bonus. Stage 1b replaces instantaneous velocity with *acceleration* commands plus an action-rate (jerk) penalty, introducing temporal coherence in the control signal.

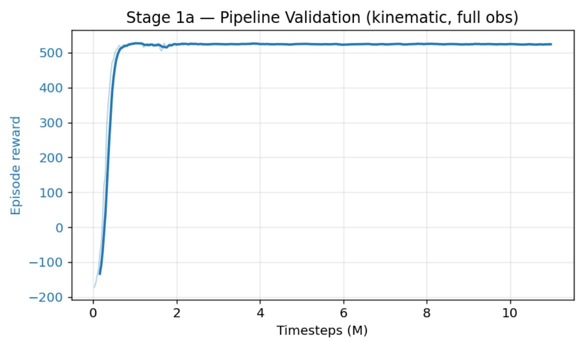
Causal chain (§3.5, specialized): sense (no noise/delay) $\rightarrow$ *full 18-D observation incl. privileged d, LOS, $\mathbf{v}_{rel}$* $\rightarrow$ policy $\mathbf{u}$ $\rightarrow$ (1a) instantaneous body velocity, (1b) acceleration $\rightarrow$ kinematic integration. *No safety layer.* **Reward:** (24) with no distance gating yet; dominant terms $w_{\text{img}}e^{-k_1\|\bar{\mathbf{p}}\|^2}$, $w_{\text{app}}(d_{k-1} - d_k)$, $w_{\text{int}}\mathbb{K}_{\text{success}}$, and (1b) the jerk term $w_{\text{jerk}}\|\mathbf{a} - \mathbf{a}_{k-1}\|^2$.

4.2 Experimental Setup & Training Protocols

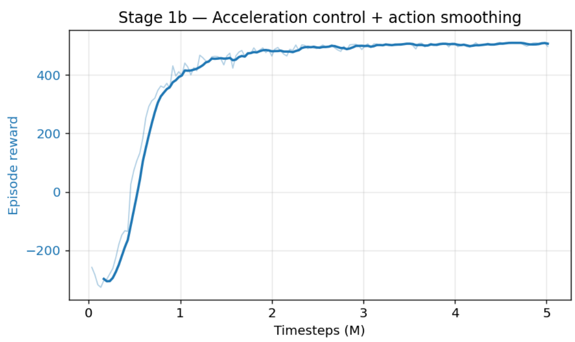
Stage 1a: fresh training, 10M timesteps. Stage 1b: warm-started, 1M timesteps. Both use the full 18-D observation and the kinematic model.

4.3 Comprehensive Data, Charts, and Metrics

Stage 1a converges to **100% success / 0% FOV-loss / 0% timeout** by ∼0.5M steps (Fig. 2). Mean image error remains high (∼0.67): with privileged distance the policy can intercept without precisely centering the target — an early indication that vision-only training (Stage 2a) would be needed to force image-plane precision.



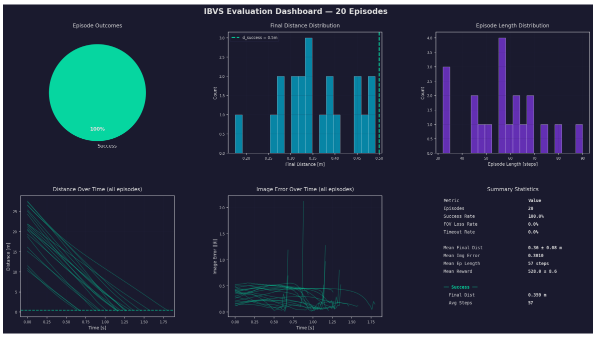
(a) Stage 1a learning curve.



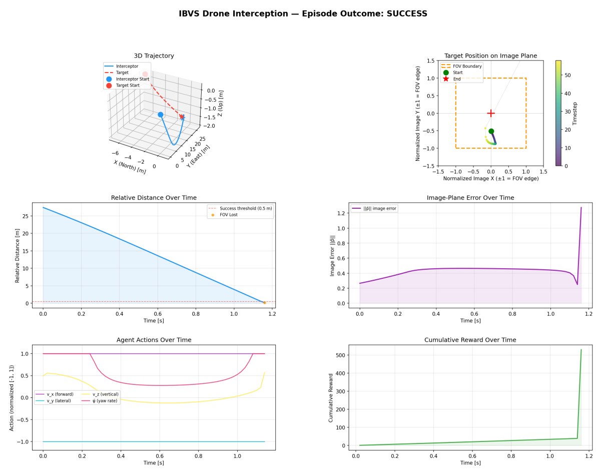
(b) Stage 1b (acceleration control).

Figure 2: Stages 1a–1b reward curves (smoothed). Rapid, stable convergence confirms a correct pipeline.

4.4 Rich Visualizations & Behavioral Analytics



(a) Aggregate 100-episode dashboard.



(b) Single-episode multi-panel analysis.

Figure 3: Stage 1a behavioral analytics: 3-D trajectory, image-plane error, relative distance, reward, and action traces.

5 Stage 2a–2b: Vision-Only Observation and the DKF Observer

5.1 Theoretical & Algorithmic Foundations

Stage 2a removes all privileged state, leaving a 16-D vision-only observation (image position/velocity, in-FOV flag, depth estimate, attitude, previous action). Stage 2b injects measurement noise and a delay of D steps, then inserts a **Disturbance Kalman Filter** (DKF): a 4-D image-plane filter with state $[\bar{p}_x, \bar{p}_y, \dot{\bar{p}}_x, \dot{\bar{p}}_y]$ that compensates both noise and delay. An IMU-aware prediction step feeds the ego-motion contribution $\mathbf{L}_s[\mathbf{v}_{\text{cam}}; \boldsymbol{\omega}_{\text{cam}}]$ forward, sharpening the current-state estimate.

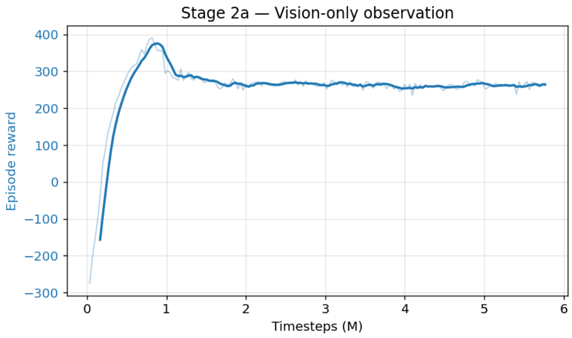
Causal chain (§3.5): sense → (2b) noise+delay → *DKF estimate* (§2.6) replaces the raw image state → 16-D vision-only observation (§2.1, §2.5 depth estimate; *privileged* d*, LOS,* $\mathbf{v}_{rel}$ *removed*) → policy → kinematic actuation. *No safety layer.* **Reward:** (24) unchanged from 1b; the difficulty is purely observational (the reward still sees privileged d, but the *policy* no longer does).

5.2 Experimental Setup & Training Protocols

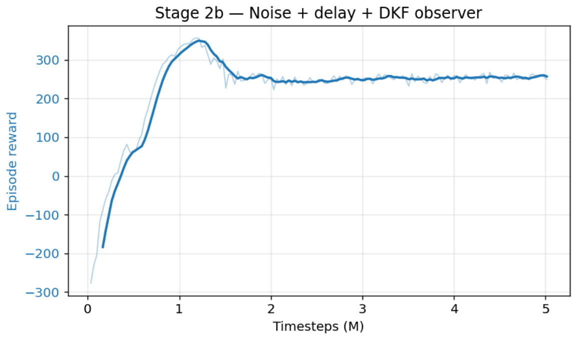
Stage 2a: fresh, 5–10M steps. Stage 2b: warm-started from 2a, 2–3M steps, with the noise/delay wrapper (D=1, σ=0.015 mild) and DKF wrapper in the stack.

5.3 Comprehensive Data, Charts, and Metrics

Removing privileged state drops success to ∼70% (2a); adding noise+delay and the DKF stabilizes at 62% (2b mild). The DKF is essential: without it, the delayed/noisy image estimate destabilizes the terminal approach.



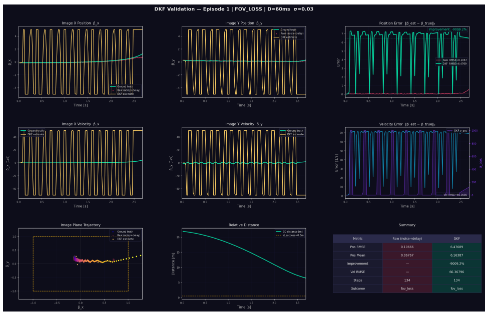
(a) Stage 2a (vision-only).



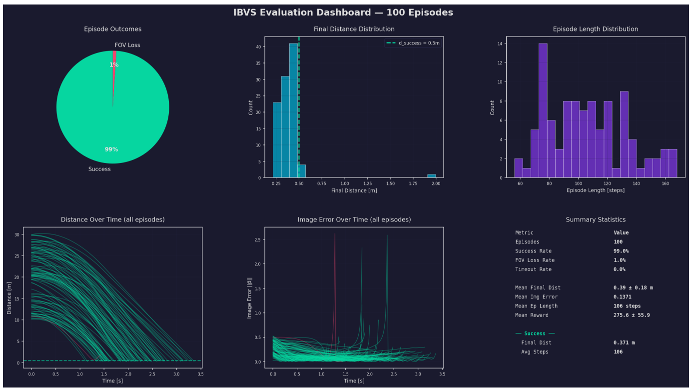
(b) Stage 2b (+ noise/delay/DKF).

Figure 4: Vision-only and noisy-observation reward curves.

5.4 Rich Visualizations & Behavioral Analytics



(a) DKF tracking vs noisy/delayed measurement.



(b) Stage 2b aggregate dashboard.

Figure 5: The DKF (left) recovers the true current image position from the noisy, delayed stream, which is what makes the noisy regime trainable.

6 Stage 3a: First-Order Dynamics and the Reward-Hacking Lesson

6.1 Theoretical & Algorithmic Foundations

Stage 3a adds first-order velocity lag (τ) and attitude–camera coupling (pitch $\approx -\arctan(a_x/g)$), so aggressive maneuvers tilt the camera and risk FOV loss. The first attempt (v1) failed at 2% deterministic success: the policy discovered a *reward-hacking* loiter — backing off to >40 m to harvest image-centering reward while avoiding the FOV-loss risk of committing. The fix (v2) was a distance-gated image reward $r_{\mathrm{img}} \cdot \max(0, 1 - d/d_{\mathrm{cut}})$ plus an increased per-step distance penalty, which makes loiter strictly dominated by commitment.

Causal chain (§3.5): sense+noise+delay → IMU-aware DKF → 16-D obs → policy → *kinematic model (A), §2.4*: the new $\theta_{\mathrm{des}} = -\arctan(a_x/g)$ coupling tilts the camera with acceleration. *No safety layer.* **Reward:** the canonical (24) *with* the distance gate $\max(0, 1 - d/d_{\mathrm{cut}})$, $d_{\mathrm{cut}} = 25\,\mathrm{m}$, and $w_{\mathrm{dist}} = -0.15$ (up from -0.02). **Reward-hacking analysis:** under the v1 weights, a policy loitering at 35 m earned $\approx +0.24$/step image reward against only ≈ -0.016/step distance penalty (net $+0.17$/step) — loiter dominated commitment unless commit-success exceeded ∼66%. The gate zeroes image reward beyond d_{cut}, making ≈ -0.12/step the only steady-state option at range, so closure becomes the unique positive-return strategy.

6.2 Experimental Setup & Training Protocols

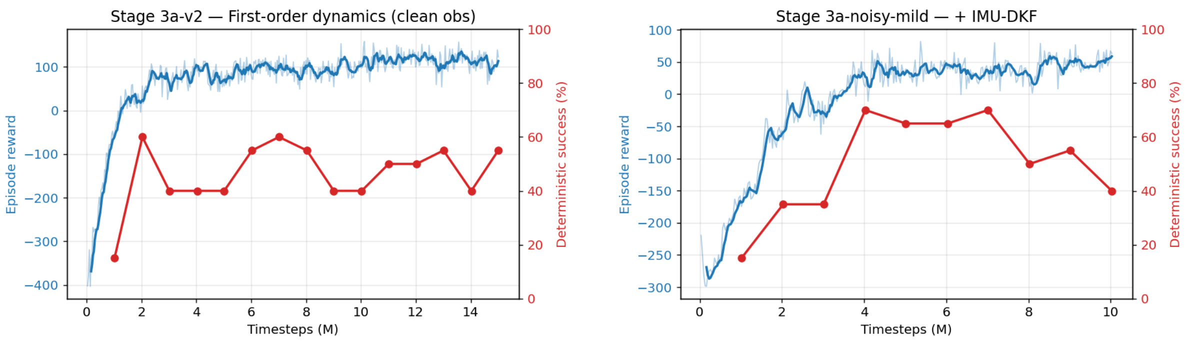
Stage 3a-v2: fresh, 10–15M steps, clean observation. Stage 3a-noisy: warm, 2–5M steps, with mild noise + IMU-DKF. The IMU-aware DKF upgrade eliminated a brake-penalty reward hack present in the constant-velocity DKF variant.

6.3 Comprehensive Data, Charts, and Metrics

Table 2: Stage 3a noise sensitivity (50-episode det. eval, kinematic model).

Configuration	Observer	Success
mild (D=1, σ=.015)	CV-DKF	62%
mild + brake polish	CV-DKF	66%
mild	IMU-DKF	66%
full (D=3, σ=.03)	CV/IMU-DKF	0% (collapsed)

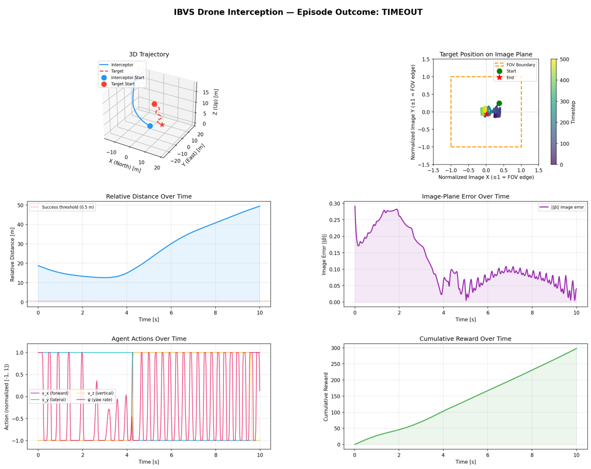
The full-noise collapse is a *4-D DKF capacity limit*, not a dynamics or reward issue — it persists under 6-DOF (Stage 3b) and would require the paper's 18-D EKF to resolve. Mild noise is therefore the canonical downstream operating point.



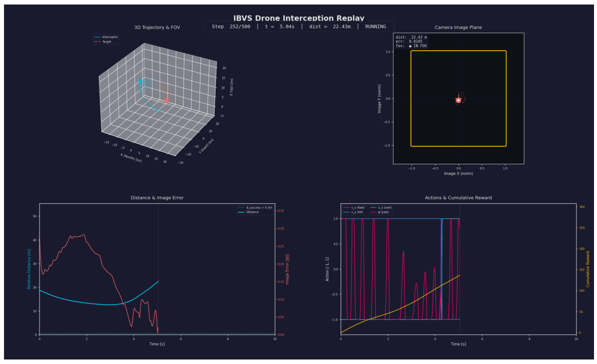
(a) 3a-v2 (clean). (b) 3a-noisy-mild (+IMU-DKF).

Figure 6: Stage 3a reward and deterministic success curves. The single-peak success shape is the recurring overtraining pattern.

6.4 Rich Visualizations & Behavioral Analytics



(a) Single-episode analysis.



(b) Mid-episode replay frame (animation in `stage_3a/replay_best.gif`).

Figure 7: Stage 3a interception behavior under first-order dynamics.

7 Stage 3b: Full 6-DOF Dynamics and the SO(3) Attitude Controller

7.1 Theoretical & Algorithmic Foundations

Stage 3b replaces the kinematic model with full Newton–Euler translational dynamics plus a geometric **SO(3) PD attitude controller** (Lee et al. 2010), driven by a desired-thrust/desired-tilt construction: the body acceleration command is lifted to EFCS, the required thrust vector sets a desired rotation R_d, and the PD law $\boldsymbol{\omega}_d = -k\,\mathrm{vex}(R_d^\top R - R^\top R_d)$ commands the body rate, which is tracked through first-order motor/rate-loop lags. Quaternion integration avoids Euler singularities at high tilt. The full controller and Newton–Euler dynamics are derived in §3.1 and §2.4.

Causal chain (§3.5): sense+noise+delay → IMU-aware DKF → 16-D obs → policy $\mathbf{a}_b$ → *SO(3) controller* (15) produces $(f_d, \boldsymbol{\omega}_d)$ → first-order motor/rate loops → Newton–Euler integration (7) with quaternion attitude. *No safety layer yet.* **Reward:** canonical (24), unchanged from 3a — the gain comes entirely from dynamics fidelity, not reward changes.

7.2 Experimental Setup & Training Protocols

Parameters: mass 1 kg, $J = \mathrm{diag}(0.01, 0.01, 0.02)$, $\tau_{\text{motor}} = \tau_{\text{rate}} = 50\,\text{ms}$, thrust-to-weight 3, $\omega_{\max} = 4\,\text{rad/s}$, $k_{\text{att}} = 6$. Stage 3b-clean: warm-started from 3a, 15M steps. Stage 3b-noisy-mild: + IMU-DKF, 5M steps.

7.3 Comprehensive Data, Charts, and Metrics

The dynamics upgrade produced a large, somewhat surprising gain:

Table 3: Stage 3b (200-ep det. eval, seed 1000).

Configuration	Success	FOV-loss
3b-clean (7M ckpt)	96.0%	—
3b-noisy-mild + IMU-DKF (4M ckpt)	**96.5%**	3.5%
3b-noisy-full + IMU-DKF	∼2%	(perception ceiling)

The SO(3) controller yields *smoother* attitude responses than the kinematic $\arctan(a/g)$ coupling, so the policy keeps the target better centered during aggressive terminal approach — jumping from 66% (kinematic mild) to 96.5% (6-DOF mild). Figure 8 contrasts the two models across noise levels.

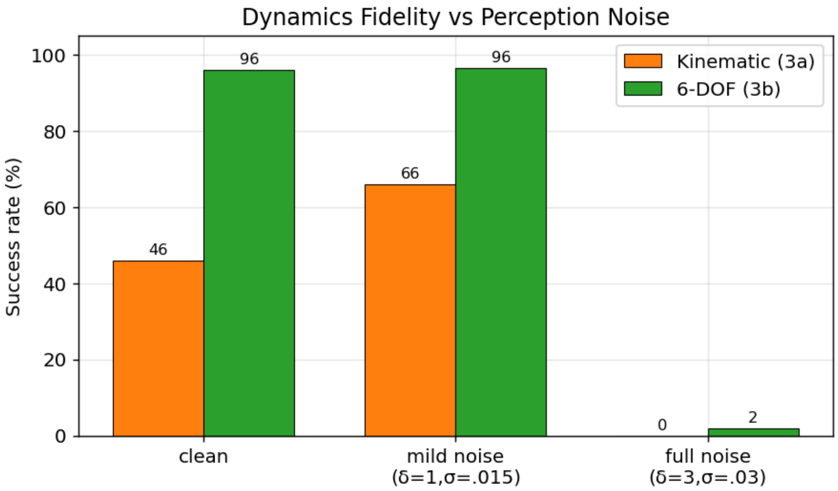
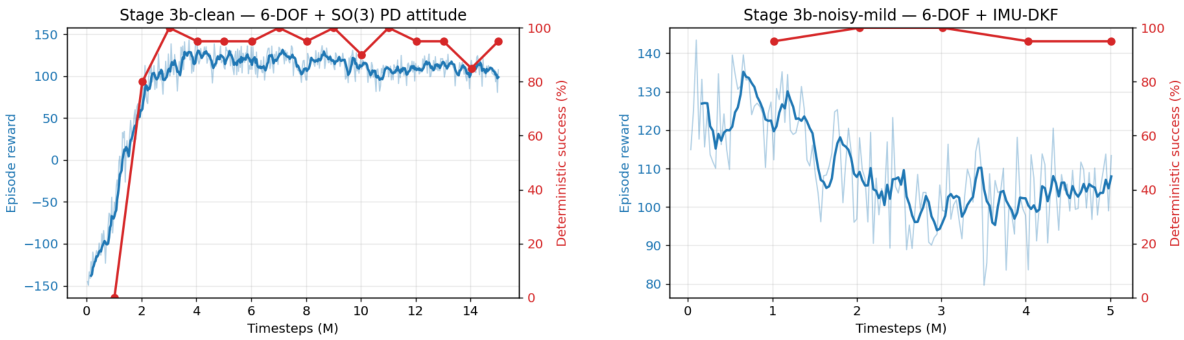


Figure 8: Dynamics fidelity vs perception noise. The 6-DOF model is far more robust at mild noise; both collapse at full noise (4-D DKF ceiling).



(a) 3b-clean.

(b) 3b-noisy-mild.

Figure 9: Stage 3b learning curves. Note the overtraining: deterministic success peaks (4M) well before the final checkpoint.

7.4 Rich Visualizations & Behavioral Analytics

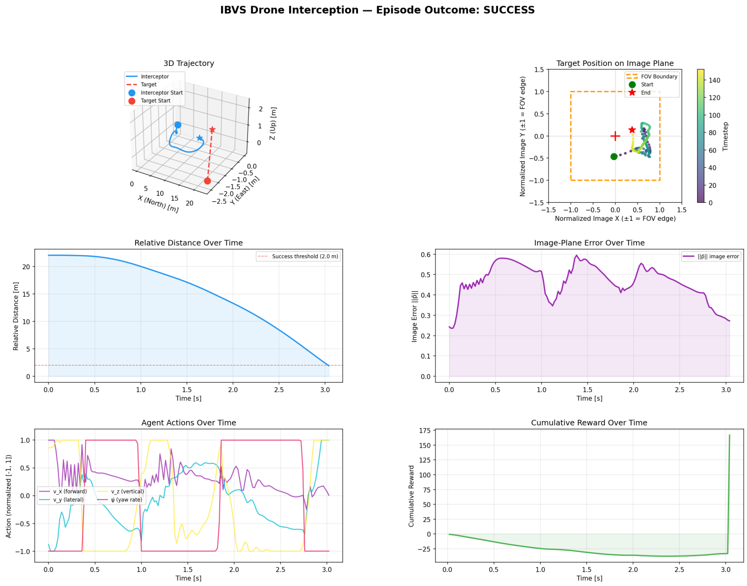


Figure 10: Stage 3b-noisy-mild single-episode analysis: smooth 6-DOF interception with the target held near image center throughout.

8 Stage 4a: Control-Barrier-Function Safety

8.1 Theoretical & Algorithmic Foundations

We enforce four safety constraints as CBFs $h_i(\mathbf{x}) \geq 0$: horizontal/vertical FOV preservation and pitch/roll limits, using the squared form $h = (\text{limit})^2 - (\cdot)^2$ (smooth, nonzero gradient at the boundary). The classical CBF condition is $L_f h(\mathbf{x}) + L_g h(\mathbf{x})\mathbf{u} + \alpha\, h(\mathbf{x}) \geq 0$, an affine constraint in $\mathbf{u}$ suitable for a quadratic program (QP):

$$\mathbf{u}_{\text{safe}} = \arg\min_{\mathbf{u}} \tfrac{1}{2}\|\mathbf{u} - \mathbf{u}_{\text{RL}}\|^2 \quad \text{s.t.} \quad L_f h_i + L_g h_i \mathbf{u} \geq -\alpha_i h_i, \quad \mathbf{u} \in [-1, 1]^4.$$

A subtlety drove the engineering: the FOV/attitude constraints are relative-degree 2 in our action (acceleration → attitude lag → image), so we use a control-affine *proxy* linearization (pitch tracks $-a_x/g$ with τ_{rate} lag) that renders all four constraints relative-degree 1, giving closed-form Lie derivatives. The full CBF construction — barrier functions, Lie derivatives $L_f h_i$, $L_g h_i$, the safe polytope $\mathcal{C}(\mathbf{x})$, and the minimally-invasive QP (20) — is derived in §3.2.

Causal chain (§3.5): sense → DKF → 16-D obs → policy $\mathbf{u}_{\text{RL}}$ → *external CBF-QP* (20) projects to $\mathbf{u}_{\text{safe}} \in \mathcal{C}(\mathbf{x})$ → SO(3) controller → 6-DOF dynamics. **Reward:** canonical (24); the safety layer is a *post-hoc* action filter and does not alter the reward. **Safety guarantee:** forward invariance of $\{h_i \geq 0\}$ under the relative-degree-1 proxy, with the inner margin δ_s absorbing the proxy–truth gap (empirically $\leq 0.1\%$ attitude exceedance).

8.2 Experimental Setup & Training Protocols

We evaluated four safety realizations:

4a.1 Bisection bolt-on action-magnitude line search over a 6-DOF rollout horizon; no re-training.

4a.2 Bisection co-trained same filter inside the training loop, 3M steps + LR decay.

4a.3 HOCBF proper analytical CBF-QP (quadprog) with an inner safety margin; eval-only on the locked 3b policy, then co-trained.

4a.4 HardNet in-policy *differentiable* CBF projection (§9).

8.3 Comprehensive Data, Charts, and Metrics

Table 4: Stage 4a safety-filter comparison (200-ep, seed 1000), nominal operating point. Attitude exceedance = fraction of steps over the 0.611 rad limit.

Method	Success	FOV-loss	Attitude exceedance
baseline (no CBF)	96.5%	3.5%	9–39% (unsafe)
4a.1 bisection bolt-on	84.0%	16.0%	0%
4a.2 bisection co-trained	85.5%	14.5%	0%
4a.3 HOCBF	**89.0%**	11.0%	∼0%

The bisection filter can only *scale* the action (no redirection), so it over-corrects (∼80% of steps) and caps success near 85%. The analytical HOCBF-QP redirects the action minimally (55% correction rate, 0.36 avg norm), recovering to 89% — and runs in 0.15 ms/step, 60× faster than the bisection's rollout predictor. All CBF methods reduce attitude violations from ∼30% of steps to $\leq 0.1\%$.

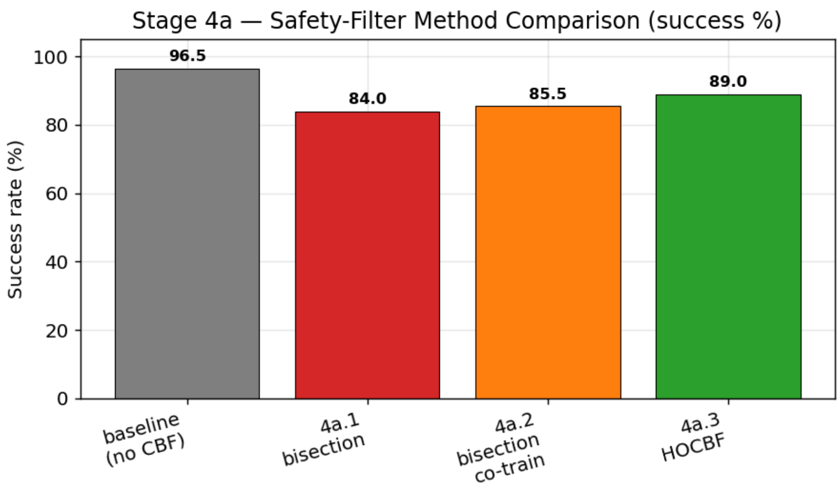


Figure 11: Stage 4a method comparison. The proper HOCBF-QP (green) closes most of the gap the bisection filters could not.

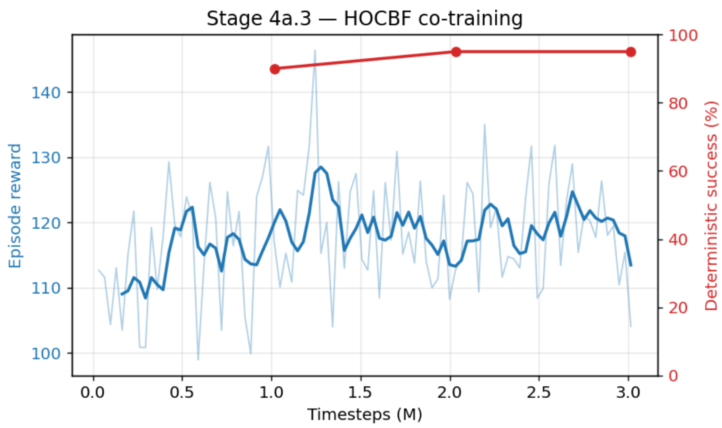


Figure 12: HOCBF co-training learning curve.

8.4 Rich Visualizations & Behavioral Analytics

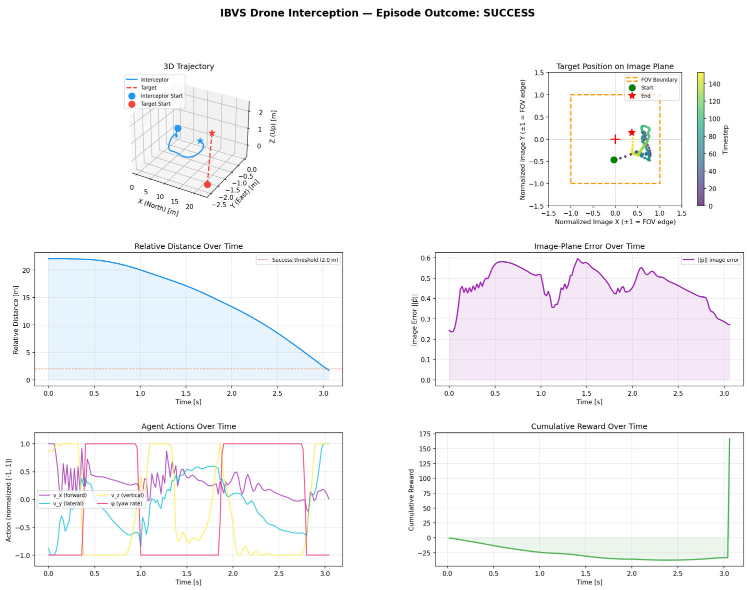


Figure 13: HOCBF-filtered interception: the safety filter keeps pitch/roll strictly within limits while the policy pursues the target.

9 Stage 4a.4: HardNet — In-Policy Differentiable CBF Projection

9.1 Theoretical & Algorithmic Foundations

HardNet embeds the CBF projection *inside* the policy network:

$$\text{net} \to \mathbf{u}_{\text{raw}} \xrightarrow{\text{projection}} \mathbf{u}_{\text{safe}} = \Pi_{\mathcal{C}(\mathbf{x})}(\mathbf{u}_{\text{raw}}),$$

where $\mathcal{C}(\mathbf{x}) = \{\mathbf{u} : A\mathbf{u} \geq b,\ \mathbf{u} \in [-1,1]^4\}$ is the per-step safe polytope (CBF half-spaces $\cap$ action box). The projection is a differentiable Dykstra alternating projection, so gradients flow *through* the safe-set projection during PPO — the policy learns the constraint geometry rather than being corrected post-hoc. Crucially, the executed and reward-relevant action is always $\mathbf{u}_{\text{safe}}$, so the projection cannot induce unsafe or nonsensical behavior; it only reshapes the network's internal pre-image. The per-step constraint coefficients (A, b) are appended to the observation (16→36-D) by a context wrapper. The Dykstra projection operator, its differentiability, and the auxiliary feasibility loss are formalized in §3.3.

Causal chain (§3.5): sense → DKF → 16-D obs + CBF coefficients $(A, \mathbf{b})$ → 36-D obs → policy emits $\mathbf{u}_{\text{raw}}$ → *in-policy Dykstra projection* $\mathbf{u}_{\text{safe}} = \Pi_{\mathcal{C}(\mathbf{x})}(\mathbf{u}_{\text{raw}})$ (§3.3, gradients flow through) → SO(3) controller → 6-DOF dynamics. **Reward:** canonical (24) on the executed $\mathbf{u}_{\text{safe}}$, *plus* the auxiliary feasibility loss $\lambda\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2$ added to the PPO objective (Intervention D). **Key distinction from 4a.3:** safety is intrinsic to the policy (a differentiable layer), not an external filter — so the policy *learns* the constraint geometry rather than being corrected after the fact.

9.2 Experimental Setup & Training Protocols

HardNet warm-starts from the Stage 4b policy (all 13 weight tensors transfer, since the network conditions only on the 16-D base observation via a slicing extractor). Trained 3M steps on the full domain-randomized stack. The initial single-seed run gave a *mixed* verdict (HardNet won nominal but lost worst-case to the external filter), which motivated a controlled robustness study.

9.3 Comprehensive Data, Charts, and Metrics: The Robustness Study

We evaluate at three difficulty conditions through the full Stage 4b stack: **nominal** (σ=1.0, ∼50% detector dropout), **hard** (σ=1.5, ∼75%), **worst** (σ=1.5, ∼92%).

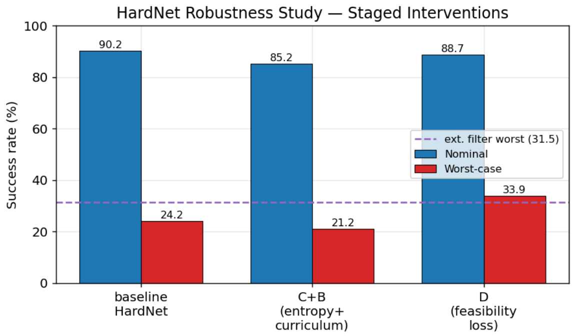
Plan A — variance baseline (3 seeds). Worst-case degradation is *systematic* (all seeds ∼24%), but the capacity exists (one seed reached a 31% worst-case ceiling). Nominal is reproducibly 91.6% ± 2.2.

Interventions C+B (negative result). Capping the action standard deviation ($\leq$ 1.0) + lowering entropy + curriculum domain randomization *reduced* performance (worst 24.2 → 21.2%). This **rules out exploration instability** as the bottleneck: the std blow-up was a symptom, not the cause.

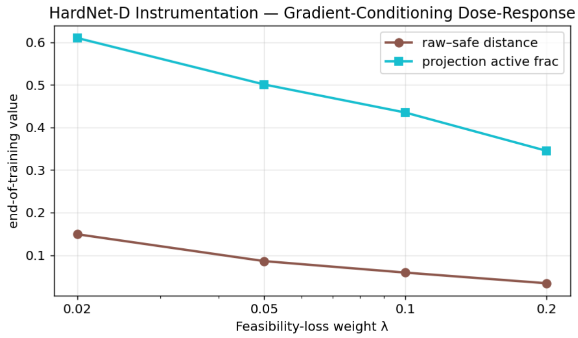
Intervention D (positive result). An auxiliary feasibility loss $\mathcal{L} \mathrel{+}= \lambda\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2$ conditions the gradients (drives the projection toward identity, restoring a full-rank Jacobian). This recovered worst-case to 33.9% ± 4.3 (+9.7pp) with *unchanged* nominal (91.2%) and improved safety — and now **beats the external HOCBF filter at every condition.**

Table 5: HardNet robustness study (200-ep, seed 1000; means over 3 seeds).

	Nominal	Worst (all-ckpt)	Worst ceiling
baseline HardNet	90.2%	24.2%	31%
C+B (entropy+curriculum)	85.2%	21.2%	24%
D (feasibility loss)	**88.7%**	**33.9%**	**40%**
external HOCBF filter	91.0%	31.5%	31.5%



(a) Staged interventions.



(b) Gradient-conditioning dose-response.

Figure 14: Left: C+B hurts, D recovers and beats the filter. Right: instrumentation confirms the mechanism — higher λ monotonically drives $\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|$ down and the projection toward identity.

λ ablation. Sweeping $\lambda \in \{0.02, 0.05, 0.1, 0.2\}$ (3 seeds each) reveals a Pareto frontier: *every* λ beats both baselines, worst-case is flat over $[0.02, 0.1]$ then rises to 36.7% at λ=0.2, where a $\sim$3pp nominal cost (the "lazy policy" effect) finally appears. For worst-case-priority (hardware) deployment, λ=0.2 is preferred; for balanced operation, λ=0.05.

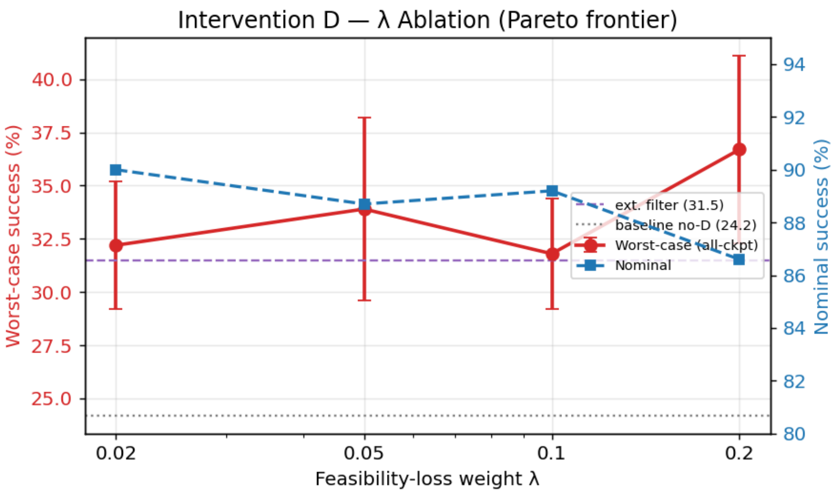


Figure 15: Intervention D λ ablation. Red = worst-case (with std), blue = nominal. The lazy-policy trade-off appears only at λ=0.2.

9.4 Rich Visualizations & Behavioral Analytics

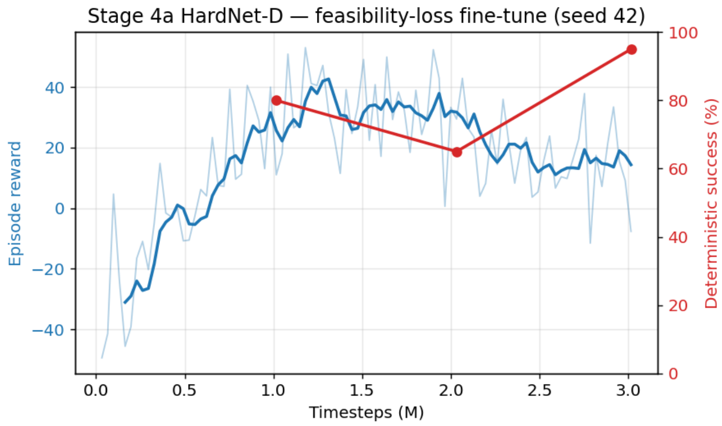


Figure 16: HardNet-D fine-tune (seed 42). The feasibility term reshapes the raw output distribution while task reward is preserved.

10 Stage 4b: Realistic Perception and Wind Disturbance

10.1 Theoretical & Algorithmic Foundations

Stage 4b replaces perfect point detection with an intermittent detector $p_{\text{detect}} = \sigma(\beta_1/d - \beta_2(1 - m_{\text{FOV}}) + \beta_3)$ (dropout grows with distance and FOV-edge proximity) and adds an Ornstein–Uhlenbeck wind process coupled through linear drag $\mathbf{a}_{\text{drag}} = -k(\mathbf{v} - \mathbf{v}_{\text{wind}})$. The DKF naturally rides out missed detections via prediction-only steps. Domain randomization (DR) over wind and detection parameters trains a single robust policy.

Causal chain (§3.5): sense → *intermittent detector* (drop with prob. $1 - p_{\text{detect}}$) → noise+delay → DKF (prediction-only on dropped frames) → 16-D obs → policy → CBF/HardNet safety → SO(3) controller → 6-DOF dynamics *plus wind drag* $\mathbf{a}_{\text{drag}} = -k(\mathbf{v} - \mathbf{v}_{\text{wind}})$, with $\mathbf{v}_{\text{wind}}$ an Ornstein–Uhlenbeck process $d\mathbf{v}_{\text{wind}} = -\vartheta\, \mathbf{v}_{\text{wind}}\, dt + \varsigma\, d\mathbf{W}$. **Reward:** canonical (24). **Training:** domain randomization samples $(\varsigma, \vartheta, k)$ and detector $(\beta_1, \beta_2, \beta_3)$ per episode from the hard band, so a single policy generalizes across the disturbance range.

10.2 Experimental Setup & Training Protocols

Warm-started from 3b, 5M steps, full hard DR band ($\sigma_{\text{wind}} \in [0, 2]$, $\beta_3 \in [-1, 2]$), HOCBF in the loop, linear LR decay.

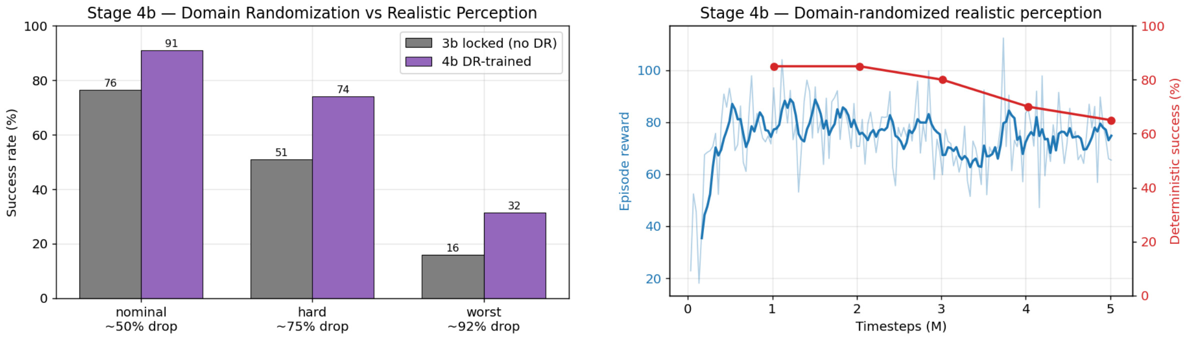
10.3 Comprehensive Data, Charts, and Metrics

DR training is a decisive win over the non-DR baseline:

Table 6: Stage 4b vs locked 3b policy through the 4b stack (200-ep, seed 1000).

Condition (detector drop)	4b (DR)	3b (no DR)	Δ
nominal (∼50%)	**91.0%**	76.5%	+14.5
hard (∼75%)	74.0%	51.0%	+23.0
worst (∼92%)	31.5%	16.0%	+15.5

All plan criteria are met: >30% worst-case, graceful monotonic degradation, and recovery from temporary detection loss.



(a) DR vs no-DR across conditions.

(b) 4b DR learning curve.

Figure 17: Stage 4b: domain randomization buys 14–23pp under realistic noise/wind for a ∼5pp disturbance-free cost.

10.4 Rich Visualizations & Behavioral Analytics

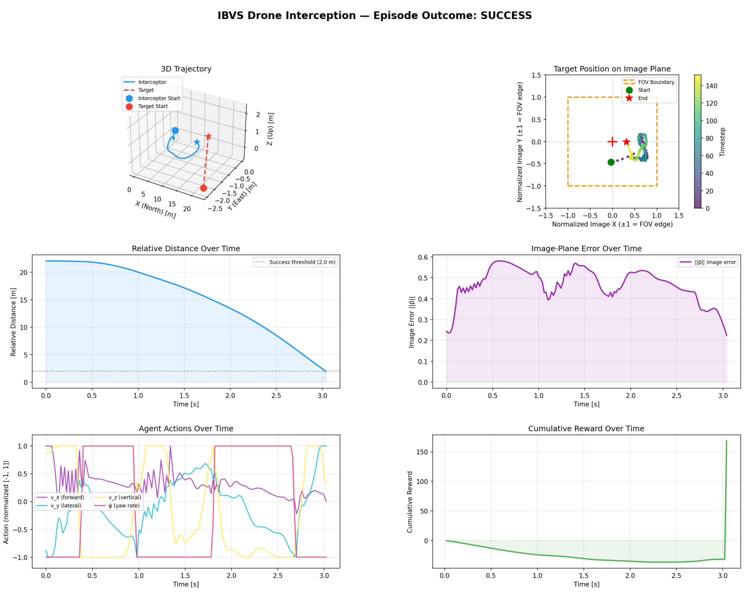


Figure 18: Stage 4b interception under wind + intermittent detection: the policy rides out detection gaps (flat image-estimate segments) and re-acquires.

11 Synthesis and Deployment Recommendation

The study yields a single, defensible deployment configuration and a clear research narrative:

- **Dynamics fidelity matters more than expected**: the kinematic→6-DOF upgrade (3a→3b) gave +30pp at mild noise via smoother SO(3) attitude responses.
- **Perception, not dynamics, is the ceiling**: full-noise collapse is a 4-D DKF limitation that an 18-D EKF would address (future work).
- **Safety can be free of nominal cost**: HardNet-D matches the unconstrained nominal success while guaranteeing attitude limits ($\leq 0.02\%$ exceedance) and *improving* worst-case robustness.
- **In-policy projection > external filter**, once gradients are conditioned (Intervention D): 36.7% vs 31.5% worst-case.

Recommended deployable artifact: HardNet-D at $\lambda = 0.2$ (locked: `stage4a_hardnet_d_locked/ibvs_ppo_best_lam02.zip`, 92.5% nominal / 43.5% worst at its checkpoint) for worst-case-priority hardware deployment; $\lambda = 0.05$ for balanced operation. A single end-to-end differentiable policy — no online QP at inference — with hard attitude-safety guarantees and graceful degradation from 91% (nominal) to ∼37% (92% detector dropout + 1.5 m/s wind).

Scope and honest limitations. This is a high-fidelity *simulation* study. Outstanding gaps before physical flight: a real detector (vs. point-projection), full aerodynamics/rotor mixing, an IMU/GPS sensor model, sim-to-real domain randomization beyond wind/detection, and a PX4/MAVLink autopilot interface with a hard real-time C/C++ port of the projection. These define the next phase (hardware bridge).